Introduction to Economics

Class 13 — Monopoly; oligopoly and the prisoner's dilemma

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Plan for today

- Monopoly: the price and quantity a single seller picks
- What a monopoly costs the rest of us
- A price for each buyer; what governments do
- Oligopoly: two sellers and a deal between them
- More sellers; when a deal can hold
- Practice: today's moves; a tax, a tariff, an external cost

Slide 1 Title

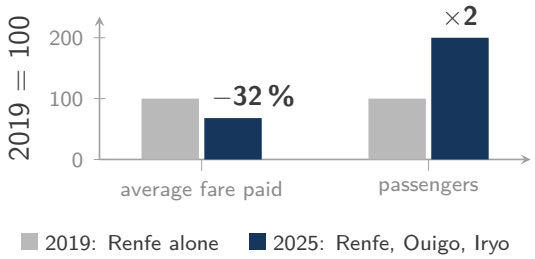
Block 4's third class, in two halves. Last Monday ended with a mountain's only cable car, a seller that chooses its own price, and left open which price it picks. The first half answers that question and measures what the answer costs everyone else; then two sellers share one market, make a deal and feel the pull to break it, and a real train line that went from one company to three shows what more sellers do to a price. The second half is practice: today's two moves as multiple-choice items, then three markets drawn on a grid, a tax, a tariff and an external cost, each worked on the board the way you would work it on paper.

Slide 2 Plan for today

The class at a glance, in two halves. Before the break, the ideas: the cable car's demand table, why each extra ride adds less to its revenue than its price, the number of rides and the price it picks, what that choice costs the riders who stay at home, a price for each rider and what governments do; then two farms that share a bay, a deal between them, why each is tempted to break it and what happens when both do, why more sellers push a price toward cost, and when a deal can hold anyway.

After the break, practice: two multiple-choice items on today's two moves, then three graph problems, a tax, a tariff and an external cost, in the form of the practice exam on the course page: you try each alone, then it is solved on the board the way you would solve it on paper.

Madrid–Barcelona by fast train: three companies instead of one — fares down 32 %, passengers doubled



until May 2021: Renfe only

10 May 2021: Ouigo starts

25 November 2022: Iryo starts

Why might more sellers push the fare down?

Comisión Nacional de los Mercados y la Competencia (CNMC), Informe anual del sector ferroviario 2025 (INF/DTSP/039/26, 30 June 2026), p. 39, table 23: Madrid–Barcelona, 2025 vs 2019; fare = nominal ticket revenue per station-to-station passenger (52.89 euros in 2025), 8.7 million passengers in 2025; entry dates: table A2, p. 56.

From the second hundred on, each adds less to revenue than its price: the price falls for all

highest price for all (euros)	13	12	11	**10**	9	**8**	7	6
rides a day	100	200	300	**400**	500	**600**	700	800
revenue a day (euros)	1,300	2,400	3,300	**4,000**	4,500	**4,800**	4,900	4,800
marginal revenue (euros a ride)	13	11	9	**7**	5	**3**	1	−1

Marginal revenue: what one more ride adds to revenue
(here: per ride, for each extra hundred)

Monopoly: the only seller of a good with no close substitute; it faces the market's demand (class 12, slide 20).
A price taker's extra bag adds just the price: marginal revenue (MR) = P = 4 (class 12, slides 6 and 8).

Slide 3 Madrid–Barcelona by fast train: three companies instead of one — fares down 32 %, passengers doubled

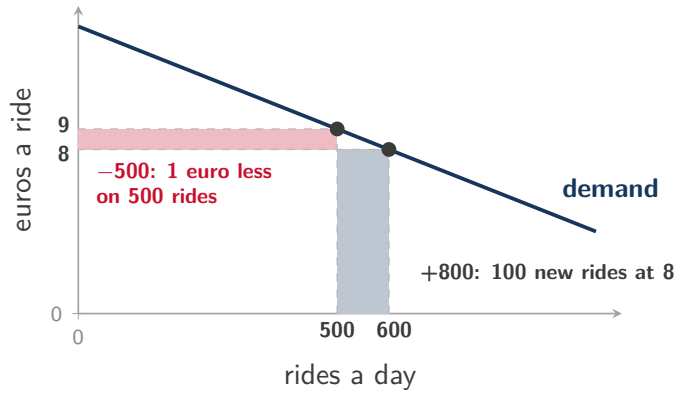
The hook: a real market where the number of sellers changed. On the high-speed line between Madrid and Barcelona one company, Renfe, sold every ticket until May 2021; Ouigo started running trains that month and Iryo in November 2022. The bars compare 2025 with 2019 and set each 2019 value to 100. The left pair is the average fare: the operators' total ticket revenue on the route divided by the number of passengers between the two stations, in euros of each year (not corrected for inflation): 52.89 euros per passenger in 2025, about a third less than in 2019; corrected for inflation the fall is larger. The right pair is the number of passengers between the two cities, which the report says doubled: 8.7 million trips in 2025. The source is the Spanish competition authority, the CNMC, in its 2025 annual report on the rail sector.

Read the figures with care: 2019 is before the pandemic, other things changed at the same time, and in 2025 alone the average fare rose while passengers fell. The arrival of two rivals is one clear reason for the cheaper fares, not the only one. The question to carry to slide 13: why might more sellers push the fare down?

Slide 4 From the second hundred on, each adds less to revenue than its price: the price falls for all

Back to last Monday's cable car, the only one up its mountain, with the same demand: for each number of rides a day, the highest price at which all of them sell, one price for every rider. Revenue is price times rides; marginal revenue is what one more ride adds to revenue, here measured per ride for each extra hundred. The gap between the top row (price) and the bottom row (marginal revenue) is the whole difference between today and last Monday's print shop: the shop could sell any number of bags at 4 euros, so one more bag added exactly the price; the cable car can sell more only by lowering the price for everyone, so each extra hundred adds less than its price, and past 700 rides it adds less than nothing. The next slide shows where the gap comes from.

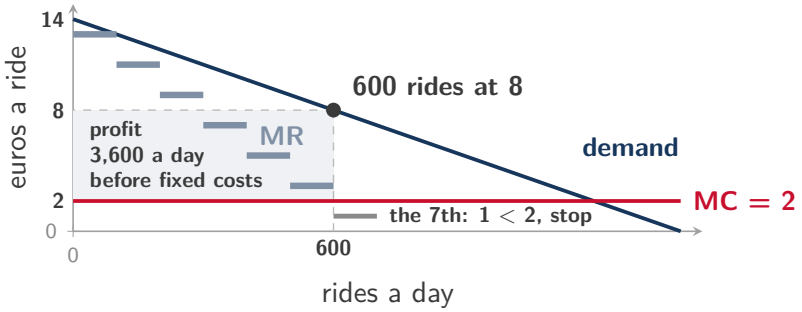
From 500 to 600 rides: 800 more from new riders, 500 less from the old ones



Marginal revenue: $800 - 500 = 300$ for 100 rides — 3 a ride, below the price of 8.

Sell every hundred whose marginal revenue beats the 2-euro cost: 600 rides at 8 euros

Each extra ride costs 2 euros (energy, wear); fixed costs are paid whatever it sells.



MR falls below MC after 600; demand sets the price: 8, far above MC; profit $(8 - 2) \times 600$.

Marginal cost (MC): what one more ride adds to total cost (class 11, slide 10). A price taker produces while MC is below the price (class 12, slide 8).

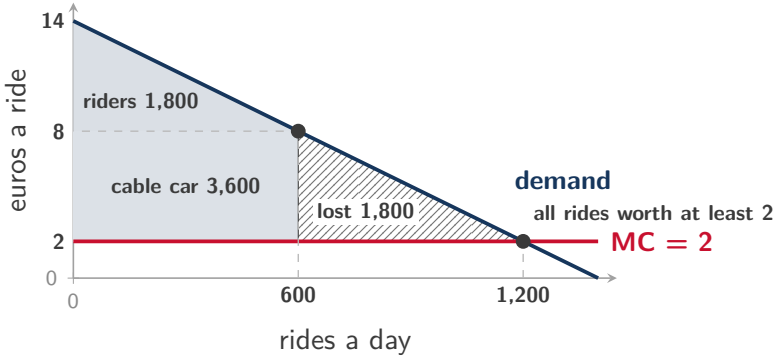
Slide 5 From 500 to 600 rides: 800 more from new riders, 500 less from the old ones

The gap in slide 4's table, drawn. Moving down the demand curve from 500 rides at 9 to 600 at 8 has two effects that pull against each other: the hundred new riders pay 8 each, but the five hundred who were already riding at 9 now pay a euro less each. Marginal revenue is what is left after the second effect, which is why it sits below the price. A price taker never feels the second effect, because it never has to lower its price to sell more; a monopoly always does. The slide assumes one price for all riders; what the cable car could do if it could charge the old riders 9 and the new ones 8 comes on slide 8.

Slide 6 Sell every hundred whose marginal revenue beats the 2-euro cost: 600 rides at 8 euros

The monopoly's choice, in two steps. New today is the cost side: each extra ride costs 2 euros, while the cables and cabins cost the same whatever it sells, so they cannot change the best number of rides (class 12's lesson on costs paid either way). Step one is the rule the print shop also followed: sell a unit as long as it adds more to revenue than to cost, and stop at the first that does not; the only change is that the revenue side is now marginal revenue, read from the steps, not the price. Step two has no counterpart for a price taker: once the quantity is fixed, the price is whatever that many riders will pay, read upward from the demand curve. Because marginal revenue is below the price, the price lands far above the 2-euro cost; the shaded box is the profit before fixed costs. Choosing the quantity with the most revenue would be wrong: revenue ignores what each ride costs.

Rides worth more than their 2-euro cost go unsold: 1,800 euros a day lost



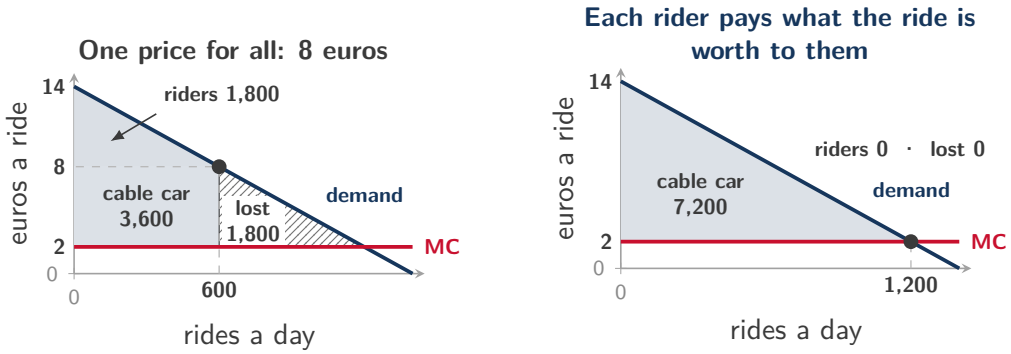
The 3,600 is a transfer from riders; the 1,800 is lost to everyone.

Consumer surplus: value minus price (class 8, slide 6) · deadweight loss: surplus lost when trades worth more than their cost do not happen (class 8, slide 17).

Slide 7 Rides worth more than their 2-euro cost go unsold: 1,800 euros a day lost

What the cable car's choice costs everyone together. A ride is worth making whenever someone values it above the 2 euros it costs, and the demand curve says what each rider would pay, so every ride up to where demand meets marginal cost should happen: 1,200 a day. At the monopoly's price only half of them do. The riders who still ride keep their surplus above the price and the cable car keeps its profit; both count as value. The riders between 600 and 1,200 value a ride above its cost but below the price, so they stay at home, and the value they would have had goes to nobody: that triangle is the deadweight loss of monopoly, the same kind of wedge a tax drove between buyers and sellers in class 8. The profit is not the loss: a euro paid by a rider is a euro received by the cable car. The loss is the rides that never happen, and it is the economic case for competition policy.

A price for each rider: all 1,200 rides sold, nothing lost — and the cable car keeps all 7,200



Price discrimination: one good, different prices for different buyers.
In practice by group: a residents' card, youth fares, booking early.

Slide 8 A price for each rider: all 1,200 rides sold, nothing lost — and the cable car keeps all 7,200

The left panel is slide 7 again, smaller: one price for all, half the worthwhile rides sold, a triangle lost. The right panel supposes the cable car knew what each rider would pay and charged exactly that. Selling one more ride then no longer lowers the price for anyone else, so the extra revenue is simply that rider's price, and the cable car sells every ride worth at least its 2-euro cost: nothing is lost, but every euro of value goes to the cable car and riders keep no surplus. No seller reads minds, so in practice buyers are sorted into groups that pay differently; this works when the groups value the ride differently and the cheap group cannot resell to the one that pays more. It explains why the same seat or ride has many prices, and why that can mean more rides sold rather than fewer. Whether buyers as a whole gain or lose depends on the case: the extra riders get a ride they would not have had, the others pay more.

Four answers to a monopoly: competition law, a price cap, public ownership — or nothing

- **Competition law:** stop mergers, fine price-fixing
- **Regulation:** cap the price a natural monopoly may charge
- **Public ownership:** the state runs it
- **Doing nothing:** when every cure costs more

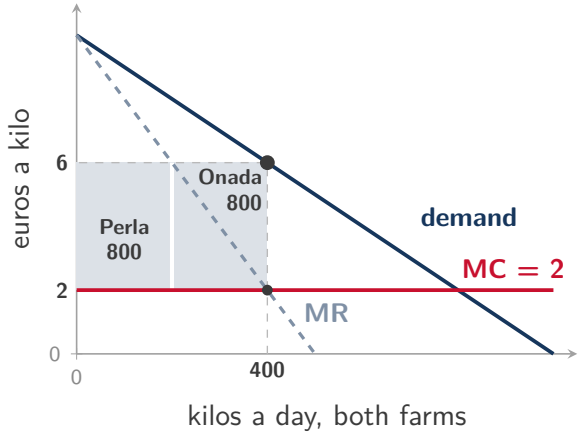
Natural monopoly: one firm serves the whole market most cheaply (class 12, slide 21).

In Spain: the CNMC (Comisión Nacional de los Mercados y la Competencia). Reading Ch. 11, pp. 258–261.

Slide 9 Four answers to a monopoly: competition law, a price cap, public ownership — or nothing

Four things a government can do about a monopoly, each with its own cost. Competition law blocks mergers that would create one and fines firms that fix prices; in Spain the body in charge is the CNMC, the same authority that wrote the train report of slide 3. Regulation caps the price a natural monopoly may charge, such as class 12's electricity grid; the cap cannot simply be set at marginal cost, because a natural monopoly's average cost lies above its marginal cost (the fixed cost is spread over the units), so at that price it would lose money; regulators allow a price near average cost, or a cap that falls year by year to keep the pressure on costs. Public ownership removes the profit motive, and with it much of the pressure to keep costs down. Doing nothing is right when the loss is small and every cure would cost more. Any news story about a regulator or a competition authority fits one of these four boxes.

Two mussel farms share one bay: acting as one monopoly, they would sell 400 kilos at 6 euros



Perla and Onada: the only two farms; each kilo costs 2 to farm.

Oligopoly: a market with a few sellers of similar goods

Cartel: sellers who agree to act as one monopoly: here 200 kilos each

Slide 10 Two mussel farms share one bay: acting as one monopoly, they would sell 400 kilos at 6 euros

From one seller to two. Perla and Onada are the only farms in the bay, so together they face the market's demand, and each kilo costs either of them 2 euros to farm. A market with a few sellers is an oligopoly, and its mark is that each seller's profit depends on what the other does. If the two acted as one firm they would use the cable car's rule: marginal revenue equals marginal cost at 400 kilos, the price read off demand is 6, and the 1,600 a day of profit splits into 800 each when each farms 200 kilos. An agreement to do this is collusion, and the group is a cartel. It pays them because together they are a monopoly, with a price far above cost: two sellers do not have to compete the price down to cost if they can agree. The next slide asks whether the agreement holds.

Selling 250 kilos instead of 200 pays each farm more — whatever the other sells

Each farm sells 200 kilos (the deal) or 250.
Price: 400 kilos → 6.00 · 450 → 5.50 · 500 → 5.00.

euros a day, Perla's first

		Onada	
		200 kilos	250 kilos
Perla	200 kilos	Perla 800, Onada 800	Perla 700, Onada 875
	250 kilos	Perla 875, Onada 700	Perla 750, Onada 750

$$\text{profit} = (\text{price} - 2) \times \text{own kilos}$$

If Onada keeps the deal, Perla earns 875 by breaking it, not 800.

Slide 11 Selling 250 kilos instead of 200 pays each farm more — whatever the other sells

The cartel's weak point, in a payoff table. Each farm picks 200 kilos (the deal) or 250; the price depends on the total, read from the demand line, and each cell shows both farms' profit a day, Perla's first, as (price minus 2) times own kilos. Compare the two cells in each column: whatever Onada does, Perla's profit is higher at 250 than at 200, and the table is symmetric, so the same holds for Onada. The reason is slide 5's price effect split between two firms: Perla's extra 50 kilos lower the price for every kilo sold, but Perla bears that cut only on its own kilos while Onada's take the rest of the hit; as one monopoly the two would count the whole cut. A promise does not change these numbers.

Both sell 250 kilos: neither gains by changing alone — yet both earn less than with the deal

euros a day, Perla's first

		Onada	
		200 kilos	250 kilos
Perla	200 kilos	Perla 800, Onada 800	Perla 700, Onada 875
	250 kilos	Perla 875, Onada 700	Perla 750, Onada 750

Dominant strategy: best whatever the other does

Nash equilibrium: each does its best, given what the other does

Prisoner's dilemma: each does its best, and both are worse off than if both had cooperated

Slide 12 Both sell 250 kilos: neither gains by changing alone — yet both earn less than with the deal

How to read any game of this kind: for each choice the other might make, circle your best reply, then look for the cell where both numbers are circled. A choice that is best whatever the other does is a dominant strategy; here each farm has one. A cell where neither player gains by changing alone is a Nash equilibrium, and this game has exactly one, both at 250. The dashed cell, the deal, pays both farms more, yet it is not where they end up, because each farm ignores what its extra kilos cost the other. A game in which each player's best choice leaves both worse off than if both had cooperated is a prisoner's dilemma, named after two suspects who both confess although both would do better staying silent. Who gains: buyers, who get 500 kilos at 5 instead of 400 at 6, with a smaller loss of surplus (450 a day instead of 800); that is why competition authorities like such dilemmas and fight cartels. A Nash equilibrium is not the best outcome for the players, only the one neither leaves on its own.

More sellers, a price closer to cost: 6, then 5, then 2 euros a kilo



Madrid–Barcelona

one company → three
average fare: −32 %
passengers: ×2

2019 → 2025
(CNMC, slide 3)

With free entry, the price falls to the lowest average total cost (class 12, slide 18).

Slide 13 More sellers, a price closer to cost: 6, then 5, then 2 euros a kilo

The hook's payoff, with the mussel farms. One seller, or two whose deal holds, sells 400 kilos at 6. Two farms each doing their best for themselves end at 500 kilos and 5. With many farms in the bay, each one's kilos barely move the price, so each sells as long as the price covers its 2-euro cost, and the price falls to cost: class 12's competitive outcome. The reason: a seller that sells more gains the price on its extra kilos and bears the price cut only on its own kilos; the smaller its share of the market, the smaller that loss and the more it wants to sell.

The card returns to slide 3: on the Madrid–Barcelona high-speed line one company became three in 2021 and 2022, and by 2025 the average fare (ticket revenue divided by passengers) was about a third lower than in 2019 while passengers had doubled, according to the Spanish competition authority, the CNMC. The same logic fits: each newcomer gains by filling its trains even if fares fall, because part of the fall lands on the others' tickets. It is one clear reason, not a proof: 2019 was before the pandemic, and new trains and other changes arrived at the same time. It is also why competition authorities count sellers when judging a merger.

Meeting every day, the deal can hold: a cheater gains 75 once, then loses 50 every day

Perla's profit so far, euros

	day 1	day 2	**day 3**
Keeps the deal (200 kilos)	800	1,600	**2,400**
Breaks it on day 1; both sell 250 from day 2	875	1,625	**2,375**

Tit-for-tat: start with the deal, then do what the other did yesterday.

In Spain, rivals' agreements on prices or quantities are banned: Ley 15/2007, art. 1.

Slide 14 Meeting every day, the deal can hold: a cheater gains 75 once, then loses 50 every day

Why some deals hold after all: the two farms do not play once but meet every morning. The table follows Perla's profit added up day by day. Keeping the deal pays 800 a day. Breaking it on day 1 pays 875 that day, 75 more; but Onada sees the extra kilos and sells 250 from day 2 on, so both earn 750 from then, 50 a day less than with the deal. The cheater is ahead by 75 after one day, by 25 after two, and behind from the third day on: the threat of tomorrow's punishment is what keeps the deal, not the promise. Tit-for-tat, start with the deal and then do what the other did yesterday, is the simplest strategy that carries this threat, and it won a famous computer tournament of repeated prisoner's dilemmas. This is why a few firms that meet again and again in one market can keep prices up without any written deal, and why such agreements are illegal: in Spain the competition law bans agreements between rivals on prices or quantities.

A monopoly sells while marginal revenue beats marginal cost and reads its price off the demand curve.

Its price is above marginal cost: rides worth more than their cost go unsold.

Each member of a cartel gains by breaking it: a prisoner's dilemma.
More sellers push the price toward marginal cost.

slide 15 of 25

Slide 15 In one line

The first half in four lines, to read slowly: the first is slides 4 to 6, the second slide 7, the third slides 10 to 12, and the fourth slides 3 and 13.

Break

Back at 11:00

slide 16 of 25

Slide 16 Break

Ten minutes. Paper copies of Problem set 3 are handed out now: your copy number is the same as on Campus Virtual. After the break: practice, two multiple-choice items and three graph problems.

Multiple choice, two minutes alone: the monopoly's price

A1 [4 points]

One firm owns the town's only fibre network. Each extra household it connects costs it 10 euros a month. The table gives the highest price at which it can connect each number of households, one price for all. Which price gives the firm the largest profit?

price (euros a month)	60	50	40	30	20	10
households (hundreds)	1	2	3	4	5	6

A 20 euros a month
B 40 euros a month
C 10 euros a month
D 30 euros a month

Answer A1

A (B) C D

B: the 3rd hundred adds 20 a household, the 4th only 0, below the cost of 10: 300 households at 40 euros

Slide 17 Multiple choice, two minutes alone: the monopoly's price

The move of slides 4 to 6 as a multiple-choice item, worked as on paper: (1) revenue of each row, price × households: 6,000, 10,000, 12,000, 12,000, 10,000, 6,000; (2) what each extra hundred adds, per household: 60, 40, 20, 0, −20, −40; (3) keep every hundred that adds more than its 10-euro cost, the first three, and stop at the fourth, which adds 0; (4) read the price from the demand row at 300 households: 40 euros, option B. Check: $(40 - 10) \times 300 = 9{,}000$ a month, more than 200 or 400 households give (8,000).

Each wrong option is a slip: 10 is price = marginal cost, the price taker's rule; 20 is the third hundred's marginal revenue read as the price, which comes from the demand row; 30 is selling while marginal revenue is not negative (revenue still at its top, 12,000 at 300 and 400), with the cost forgotten. A wrong answer costs nothing: never leave one blank.

Multiple choice, two minutes alone: open on Sundays?

A2 [4 points]

A town's only two supermarkets, Fresc and Rodó, each choose whether to open on Sundays. Profit in thousand euros a month, Fresc's first:

		Rodó closed	**Rodó** open
Fresc	closed	Fresc 12, Rodó 10	Fresc 8, Rodó 13
	open	Fresc 15, Rodó 6	Fresc 9, Rodó 8

Which outcome is the Nash equilibrium?

A Fresc: closed; Rodó: closed.
B Fresc: open; Rodó: closed.
C Fresc: closed; Rodó: open.
D Fresc: open; Rodó: open.

Answer A2

A B C (D)

D: opening is best for each, whatever the other does: Fresc 15 > 12 and 9 > 8; Rodó 13 > 10 and 8 > 6

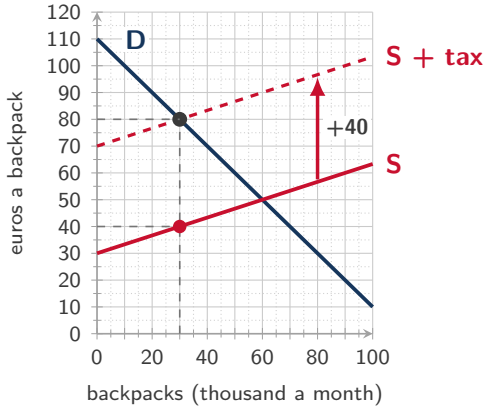
Slide 18 Multiple choice, two minutes alone: open on Sundays?

Slide 12's method on a new table, as a multiple-choice item: for each choice of the other shop, circle a shop's own best reply, comparing only its own two numbers. Fresc: if Rodó stays closed, 15 beats 12; if Rodó opens, 9 beats 8: open either way. Rodó: 13 beats 10 and 8 beats 6: open either way. The cell where both numbers are circled, both open with 9 and 8, is the Nash equilibrium, option D; opening is a dominant strategy for each, and both would earn more closed (12 and 10): a prisoner's dilemma.

The wrong options: A is the cell with the largest total, what both would like, not where they end up; B and C take one shop's largest number without asking what the other does. A shop never compares its number with the other's.

A tax on backpacks: the line and the prices — three minutes alone

The grid shows the market for backpacks: demand D and supply S. The state puts a tax of 40 euros on every backpack sold; the sellers pay it to the state.



[5 points; drawing 2; boxes as marked]

(a) Draw the supply curve with the tax: at every quantity 40 euros above S. Label it S + tax.

(b) With the tax (quantity in thousand, prices in euros):

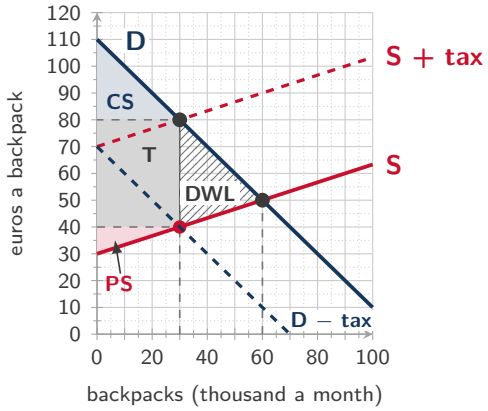
Quantity sold [1] **30 thousand**

Buyers pay (euros a backpack) [1] **80 euros**

Sellers keep (euros a backpack) [1] **80 − 40 = 40 euros**

Draw on the grid and label every line you draw; numbers in the boxes, working beside them.

The tax's areas: numbers, and letters inside — five minutes alone



[9 points; letters 2; boxes as marked]

(c) With the tax of 40 euros a backpack, in thousand euros a month: CS, PS, T and DWL (below). Write each one's letters inside its area.

CS [1.5] **CS = 450**

PS [1.5] **PS = 150**

T [1.5] **T = 1,200**

DWL [1.5] **DWL = 600**

(d) Who bears more of the tax? Circle one. [1]

(Buyers) (Sellers) buyers pay 30 more (50 → 80), sellers keep 10 less (50 → 40)

(e) Same tax, but the buyers pay it to the state: which line would you draw? Would any box change?

D − tax, 40 below D: it meets S at 30 thousand; sellers keep 40, buyers pay 80 a backpack — the same answers

CS: consumer surplus · PS: producer surplus · T: tax revenue · DWL: deadweight loss. An area's letters go inside it (a small area: beside it, with an arrow). A triangle is ½ × base × height.

Slide 19 A tax on backpacks: the line and the prices — three minutes alone

Worked as you would write it on paper. (1) Read the scale first: one solid square is 10 (10 thousand across, 10 euros up), the dotted lines are the 5s; other grids count differently, so always read the axis. (2) The line: the sellers hand over 40 a backpack, so at every quantity they need 40 more: take two points of S, (0, 30) and (60, 50), add 40 to each, (0, 70) and (60, 90), draw the line through them, parallel to S, and label it S + tax.

(3) Quantity sold: where S + tax meets D, 30 thousand (check: 110 − 30 = 80 = 70 + 10). (4) Buyers pay the crossing's height, 80 euros a backpack. (5) Sellers keep 80 − 40 = 40 euros, the point on S straight below the crossing. Write the working beside each box: a bare number earns partial points or none.

Slips that cost points: a steeper line from S's own start, no label, the sellers' price read at the crossing (80 is what buyers pay), the quantity read where S meets D (60 thousand is the market before the tax).

Slide 20 The tax's areas: numbers, and letters inside — five minutes alone

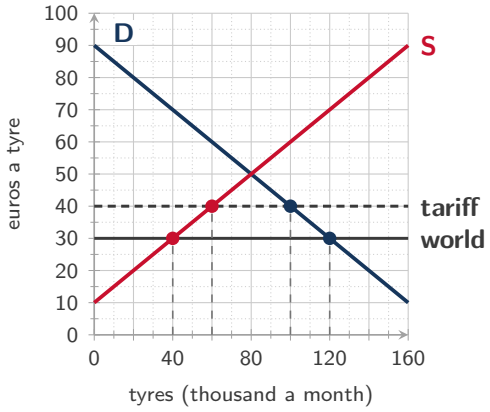
Thousand euros a month. CS: under D, above the 80 buyers pay, out to the 30 sold: ½ × 30 × 30 = 450, CS inside. PS: above S, below the 40 sellers keep, out to 30: ½ × 30 × 10 = 150; a thin triangle, one square high, so its letters go beside it with an arrow. T: tax × quantity sold, 40 × 30 = 1,200, the rectangle between the two prices. DWL: the backpacks from 30 to 60 thousand, worth more to buyers than they cost sellers and no longer sold: the triangle between D and S, its side at 30 the tax, 40, its width 60 − 30 = 30: ½ × 40 × 30 = 600. Check: the four add to 2,400, the surplus without the tax.

(d) Buyers pay 30 more (50 → 80), sellers keep 10 less (50 → 40): buyers bear more, because demand is the steeper side; the side that cannot walk away pays more, whoever hands the tax over. (e) If the buyers paid it, draw D − tax, 40 below D: it meets S at the same 30 thousand, sellers keep 40 and buyers pay 80 a backpack in all: the same answers.

Slips: CS from 0 instead of from the buyers' price; T as a triangle or as tax × 60; DWL between S + tax and S; an area without its letters.

Tyres: opening to trade, then a tariff — four minutes alone

The grid shows the market for tyres in a country that does not trade: demand D and supply S.



[6.5 points; drawing 2; boxes as marked]

(a) The country opens to trade at a world price of 30 euros a tyre. Draw it as a line; label it "world".

Production [1] | Consumption [1] | Imports [0.5]

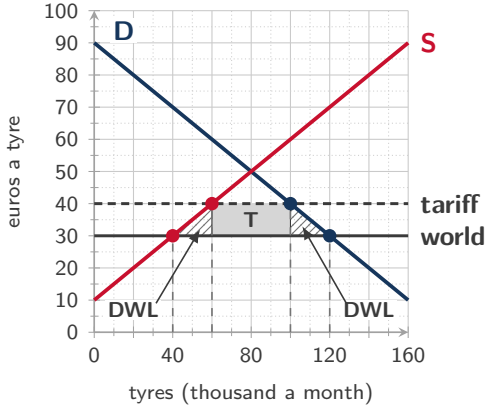
Production [1]	Consumption [1]	Imports [0.5]
40 thousand	**120 thousand**	**120 − 40 = 80**

(b) Then a tariff of 10 euros on every imported tyre. Draw the price with the tariff; label it "tariff".

Price [0.5]	Production [0.5]
30 + 10 = 40 euros	**60 thousand**

Consumption [0.5]	Imports [0.5]
100 thousand	**100 − 60 = 40**

The tariff's areas: T and two DWL triangles — three minutes alone



[5 points; letters 1; boxes as marked]

(c) With the tariff (world price 30, tariff 10), in thousand euros a month: the tariff revenue (T) and the deadweight loss (DWL, both triangles together). Write T inside its area, and DWL beside each of its two triangles, with an arrow.

T [2]

T [2]
T = 10 × 40 = 400

DWL (total) [2]

DWL (total) [2]
DWL = 100 + 100 = 200

Thousand tyres × euros = thousand euros. A triangle is ½ × base × height.

Slide 21 Tyres: opening to trade, then a tariff — four minutes alone

Worked as on paper. (1) Scale: across, one solid square is 20 thousand tyres and the dotted lines are the 10s; up, 10 euros a square, dotted 5s. Without trade the market clears at 80 thousand and 50 euros. (2) Trade: at 30 the country can buy as many tyres as it likes abroad, so draw a flat line at 30 across the grid and label it world. Production is read on S at 30: 40 thousand; consumption on D at 30: 120 thousand; imports are the gap, 120 − 40 = 80.

(3) Tariff: an imported tyre now costs 30 + 10 = 40 in the country, so draw a second flat line at 40, labelled tariff, and read again: production on S, 60 thousand; consumption on D, 100 thousand; imports 100 − 60 = 40. Slips: the tariff alone (10) in the price box instead of world price plus tariff; production read on D and consumption on S; imports given as the no-trade quantity.

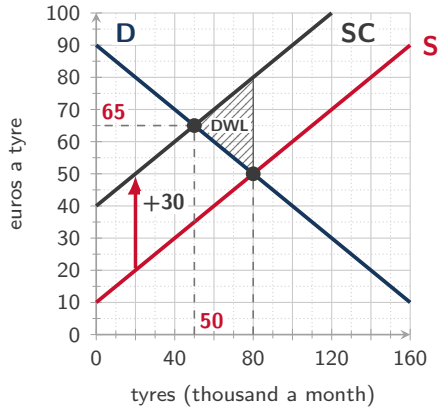
Slide 22 The tariff's areas: T and two DWL triangles — three minutes alone

Thousand tyres × euros = thousand euros. T: the tariff times the imports with the tariff, 10 × 40 = 400: the rectangle between 30 and 40 from 60 to 100 thousand, with T written inside it. The slip: 10 × 80, the imports before the tariff. DWL, left triangle: the tyres from 40 to 60 thousand are now made at home at a cost above the world price: under S, above 30, from 40 to 60, ½ × 20 × 10 = 100. Right triangle: the buyers from 100 to 120 thousand who valued a tyre above 30 but below 40 now go without: ½ × 20 × 10 = 100. Total 200.

These triangles are small, so the letters go beside each with an arrow into it; T fits inside its rectangle. The letters are points of their own and the quickest on the page. Slips: one triangle only; the triangles taken between S and D (the no-trade picture); T as the whole rectangle up to the no-trade price.

The same tyres with smoke: an external cost — four minutes alone

The tyre market without trade, D and S as before. Making each tyre also sends out smoke that harms people nearby: 30 euros of harm per tyre, which buyers and sellers do not count.



[10 points; drawing 4; boxes as marked]

(a) Draw the social-cost curve (SC: the sellers' cost plus the harm to others); label it SC.

(b) Market quantity [1] Efficient quantity [1.5]

80 thousand | **50 thousand**

(c) The deadweight loss (DWL) at the market quantity, thousand euros a month; label its area DWL.

(d) The tax per tyre that brings the market to the efficient quantity.

(c) DWL [2] (d) Tax (euros a tyre) [1.5]

DWL = 450 | **30 euros a tyre: the harm**

Hoodies: from costs to a choice — four minutes alone (optional)

[8 points; boxes as marked]

A workshop sews hoodies and sells them at the market price. Its total cost, euros a week:

hoodies a week	0	25	50	75	100
total cost (euros)	400	700	900	1,300	1,900
marginal cost, MC (euros a hoodie)		**12**	**8**	**16**	**24**

(a) Fill in the marginal cost (MC) of each extra 25 hoodies, per hoodie. [2]

(b) At 50 hoodies: the average variable cost (AVC)? The average total cost (ATC)? [2]
AVC = (900 − 400) ÷ 50 = 10; ATC = 900 ÷ 50 = 18

(c) The price is 20 euros a hoodie. How many hoodies? Profit? [2]
75 hoodies (MC 16 < 20 < 24): profit 1,500 − 1,300 = 200 euros

(d) The price falls to 14. Produce or shut down? Why? [2]
produce 50: it loses 200 (700 − 900), less than the fixed cost of 400 it pays if it shuts (lowest price to produce: 10, the lowest AVC)

Variable cost = total cost − fixed cost (class 11, slides 9–11).
A price taker produces while MC < price (class 12, slide 8).

Slide 23 The same tyres with smoke: an external cost — four minutes alone

(a) The harm is 30 for every tyre, whatever the quantity, so the social cost is S moved up by 30 at every quantity: parallel to S, through (0, 40) and (60, 70), labelled SC. (b) Market quantity: S meets D, 80 thousand, because buyers and sellers count only their own values and costs. Efficient quantity: where SC meets D, 50 thousand at 65 euros ($90 - Q/2 = 40 + Q/2$), on the dotted lines at 50 and 65.

(c) The tyres from 50 to 80 thousand cost society (SC) more than buyers value them (D): the triangle between SC and D from 50 to 80; its vertical side at 80 runs from D (50) to SC (80), 30 = the harm, and its width is 80 − 50 = 30: ½ × 30 × 30 = 450, with DWL inside. (d) A tax equal to the harm, 30 a tyre: the sellers then face SC as their cost and the market stops at 50. Same move as slide 19's tax, read the other way: there a tax moved supply up, here the harm does, and the right tax makes the market feel it.

Slips: SC steeper than S or from the origin; the efficient quantity where SC meets S; the triangle between S and D or between SC and S; the tax read as 15, the rise in what buyers pay.

Slide 24 Hoodies: from costs to a choice — four minutes alone (optional)

An optional problem in the firm's form, from a cost table to the choice. (a) Each extra 25 hoodies adds 300, 200, 400, 600 to the cost: per hoodie 12, 8, 16, 24. (b) The cost at 0 hoodies, 400, is the fixed cost; at 50 the variable cost is 900 − 400 = 500, so AVC = 500 ÷ 50 = 10 and ATC = 900 ÷ 50 = 18. (c) Make each batch whose MC is below the price of 20: 12, 8 and 16 are, 24 is not: 75 hoodies; profit 20 × 75 − 1,300 = 200 (50 or 100 hoodies give 100).

(d) At 14 the batches with MC 12 and 8 pay: 50 hoodies, revenue 700, cost 900, a loss of 200; shutting down still pays the fixed cost, a loss of 400: produce. The rule compares the price with the average variable cost (lowest 10, at 50), never with ATC: the fixed cost is paid either way. Slips: "14 < 18, shut down"; AVC with the fixed cost in it (900 ÷ 50); the MC of a batch not divided by 25.

Before Thursday 12 November

- Thursday 12 November: monopolistic competition; problems on Blocks 3–4 — read Mankiw & Taylor Ch. 12, pp. 267–276
- Problem set 3: paper copy handed out today; bring your worked solutions and your copy on Thursday (not handed in)
- Practise in Sextant: the units the course page lists

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Slide 25 Before Thursday 12 November

Thursday 12 November: monopolistic competition, many sellers each with its own slightly different product, and problems on Blocks 3 and 4. Reading: Mankiw & Taylor chapter 12, pages 267 to 276. Problem set 3, on Block 4, is not handed in: paper copies of Problem set 3 were handed out today, and your copy number is the same as on Campus Virtual; work it on your own paper and bring your worked solutions, with your copy of the set, to class on Thursday 12 November. Its problems 4 and 5 use today's two first-half moves, the demand table and the two-by-two game. For the graphs, the practice exam on the course page: do it in one sitting; its sample solution follows there. Practise in Sextant with the units the course page lists.

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